19523

MASTERING HTML, CSS & JavaScript Web Publishing

Laura Lemay
Rafe Colburn
Jennifer Kyrnin

RUNGTA COLLEGE OF ENGINEERING & TECHNOLOGY
LIBRARY
KOHKA KUR... BHILAI

RUNGTA COLLEGE OF ENGG & TECH.
Acc. No. 19523
Date. 22/11/12
KOHKA-KURUD.BHILAI

BPB PUBLICATIONS
20 Ansari Road, Darya Ganj, New Delhi-110002

Book Bank

Contents at a Glance

Introduction 1

PART I: Getting Started

1 What Is Web Publishing? 7
2 Getting Your Tools in Order 23
3 Introducing HTML and CSS 41

PART II: Creating Web Pages

4 Learning the Basics of HTML 57
5 Organizing Information with Lists 71
6 Working with Links 89

PART III: Doing More with HTML and CSS

7 Formatting Text with HTML and CSS 121
8 Using CSS to Style a Site 159
9 Using Images on Your Web Pages 197
10 Building Tables 241
11 Using CSS to Position Elements on the Page 287
12 Designing Forms 313
13 Structuring a Page with HTML5 371
14 Integrating Multimedia: Video and Sound 385
15 Advanced CSS: Page Layout in CSS 421
16 Using Responsive Web Design 443

PART IV: Using JavaScript and jQuery

17 Introducing JavaScript 471
18 Using jQuery 499
19 Using JavaScript in Your Pages 527
20 Working with Frames and Linked Windows 555

PART V: Designing for Everyone

21 Designing for the Mobile Web 575
22 Designing for User Experience 605

PART VI: Going Live on the Web

23 How to Publish Your Site 625
24 Taking Advantage of the Server 651
25 Search Engines and SEO 687
Index 703

Table of Contents

Introduction 1

PART I: Getting Started

LESSON 1: What Is Web Publishing? 7

Thinking Like a Web Publisher 8

The Web Is a Hypertext Information System 8

The Web Is Cross-Platform 9

The Web Is Distributed 9

The Web Is Dynamic 10

The Web Is Interactive 12

Web Browsers 14

What the Browser Does 14

An Overview of Some Popular Browsers 15

Web Servers 18

Uniform Resource Locators 18

Defining Web Publishing Broadly 19

Summary 20

Workshop 20

Q&A 20

Quiz 21

Quiz Answers 21

Exercises 21

LESSON 2: Getting Your Tools in Order 23

Anatomy of a Website 24

Setting Up Your Computer for Web Publishing 26

Text Editors 26

A Web Browser 28

Using the Google Chrome Developer Tools 29

What Do You Want to Do on the Web? 33

Wireframing Your Website 33

What's Wireframing, and Why Do I Need It? 34

Hints for Wireframing 35

Web Hosting ... 36
Using a Content-Management Application ... 36
Setting Up Your Own Web Hosting ... 37
Summary ... 38
Workshop ... 39
Q&A ... 39
Quiz ... 39
Quiz Answers ... 40
Exercises ... 40

LESSON 3: Introducing HTML and CSS **41**

What HTML Is (And What It Isn't) ... 42
HTML Describes the Structure of a Page ... 42
HTML Does Not Describe Page Layout ... 43
Why It Works This Way ... 44
How Markup Works ... 45
What HTML Files Look Like ... 45
Text Formatting and HTML ... 49
HTML Attributes ... 50
Using the `style` Attribute ... 50
Including Styles in Tags ... 51
A Short History of HTML Standards ... 52
XHTML ... 52
The Current and Evolving Standard: HTML5 ... 53
Summary ... 54
Workshop ... 54
Q&A ... 54
Quiz ... 54
Quiz Answers ... 55
Exercises ... 55

PART II: Creating Web Pages

LESSON 4: Learning the Basics of HTML **57**

Structuring Your HTML ... 58
The `<html>` Tag ... 58
The `<head>` Tag ... 59
The `<body>` Tag ... 59

The Title 60
Headings 62
Paragraphs 64
Comments 65
Summary 68
Workshop 68
Q&A 68
Quiz 69
Quiz Answers 69
Exercises 70

LESSON 5: Organizing Information with Lists 71

Lists: An Overview 72
Numbered Lists 73
Customizing Ordered Lists 74
Unordered Lists 78
Customizing Unordered Lists 78
Definition Lists 81
Nesting Lists 82
Other Uses for Lists 84
Summary 85
Workshop 87
Q&A 87
Quiz 88
Quiz Answers 88
Exercises 88

LESSON 6: Working with Links 89

Creating Links 90
The Link Tag: <a> 90
Linking Local Pages Using Relative and Absolute Pathnames 95
Absolute Pathnames 96
Should You Use Relative or Absolute Pathnames? 97
Links to Other Documents on the Web 98
Linking to Specific Places Within Documents 104
Creating Links and Anchors 104
The name Attribute of the <a> Tag 105
Linking to Elements in the Same Document 110

Anatomy of a URL 110
Parts of URLs 110
Special Characters in URLs 112
The `rel` Attribute 113
Kinds of URLs 113
HTTP 113
Anonymous FTP 114
Non-Anonymous FTP 115
Mailto 115
File 116
Summary 117
Workshop 117
Q&A 118
Quiz 119
Quiz Answers 120
Exercises 120

PART III: Doing More with HTML and CSS

LESSON 7: Formatting Text with HTML and CSS 121

Character-Level Elements 122
Semantic HTML Tags 122
Changes to Physical Style Tags in HTML5 124
Character Formatting Using CSS 125
The Text Decoration Property 125
Font Properties 126
Preformatted Text 128
Horizontal Rules (or Thematic Breaks) 130
Attributes of the `<hr>` Tag 132
Line Break 133
Addresses 134
Quotations 135
Special Characters 138
Character Encoding 139
Character Entities for Special Characters 140
Character Entities for Reserved Characters 141

Fonts and Font Sizes ... 142
Summary ... 154
Workshop ... 156
Q&A ... 156
Quiz ... 157
Quiz Answers ... 157
Exercises ... 158

LESSON 8: Using CSS to Style a Site ... **159**

Including Style Sheets in a Page ... 160
Creating Page-Level Styles ... 160
Creating Sitewide Style Sheets ... 161
Selectors ... 162
Contextual Selectors ... 162
Classes and IDs ... 163
What Cascading Means ... 165
Units of Measure ... 166
Specifying Colors ... 168
Editing Styles with Developer Tools ... 171
Using Color ... 172
Links ... 173
The Box Model ... 174
Borders ... 175
Margins and Padding ... 177
Controlling Size and Element Display ... 182
Float ... 185
More Selectors ... 189
Pseudo-Classes ... 190
Attribute Selectors ... 193
The `<body>` Tag ... 194
Summary ... 194
Workshop ... 195
Q&A ... 195
Quiz ... 196
Quiz Answers ... 196
Exercises ... 196

LESSON 9: Using Images on Your Web Pages 197

Images on the Web 198
Image Formats 198
GIF 198
JPEG 199
PNG 199
SVG 200
Inline Images in HTML: The `<img>` Tag 200
Adding Alternative Text to Images 201
Images and Text 205
Text and Image Alignment 207
Wrapping Text Next to Images 210
Adjusting the Space Around Images 213
Images and Links 214
Other Neat Tricks with Images 219
Image Dimensions and Scaling 219
Image Backgrounds 220
Using Images as Bullets 224
What Is an Imagemap? 226
Getting an Image 226
Determining Your Coordinates 227
The `<map>` and `<area>` Tags 230
The `usemap` Attribute 231
Image Etiquette 237
Summary 237
Workshop 238
Q&A 238
Quiz 239
Quiz Answers 239
Exercises 240

LESSON 10: Building Tables 241

Creating Tables 242
Table Parts 242
The `<table>` Element 243
Summarizing the Table 243
Rows and Cells 244

Empty Cells ... 246
Captions ... 247
Sizing Tables, Borders, and Cells ... 251
Setting Table Widths ... 252
Changing Table Borders ... 253
Cell Padding ... 256
Cell Spacing ... 257
Column Widths ... 259
Table and Cell Color ... 260
Aligning Your Table Content ... 262
Table Alignment ... 263
Cell and Caption Alignment ... 264
Spanning Multiple Rows or Columns ... 266
More Advanced Table Enhancements ... 276
Grouping and Aligning Columns ... 276
Grouping and Aligning Rows ... 279
How Tables Are Used ... 282
Summary ... 282
Workshop ... 284
Q&A ... 285
Quiz ... 285
Quiz Answers ... 285
Exercises ... 286

LESSON 11: Using CSS to Position Elements on the Page ... 287

Positioning Schemes ... 288
Relative Positioning ... 288
Absolute Positioning ... 293
Positioning Properties ... 293
Positioning Properties and Height and Width ... 294
Nesting Absolutely Positioned Elements ... 295
Dynamic Overlays ... 297
Fixed Positioning ... 301
Controlling Stacking ... 303
Creating Drop-Down Menus ... 306
Summary ... 311

Workshop ... 311
Q&A ... 311
Quiz ... 312
Quiz Answers ... 312
Exercises ... 312

LESSON 12: Designing Forms ... 313

Understanding Form and Function ... 314
Using the `<form>` Tag ... 319
Using the `<label>` Tag ... 323
Creating Form Controls with the `<input>` Tag ... 324
Creating Text Controls ... 325
Adding Options to Text Fields with `datalist` ... 327
Using the New HTML5 Controls ... 328
Creating Password Controls ... 332
Creating Submit Buttons ... 333
Creating Reset Buttons ... 334
Creating Check Box Controls ... 334
Creating Radio Buttons ... 336
Using Images as Submit Buttons ... 337
Creating Generic Buttons ... 337
Hidden Form Fields ... 338
The File Upload Control ... 339
Using Other Form Controls ... 340
Using the `button` Element ... 340
Creating Large Text-Entry Fields with `textarea` ... 341
Creating Menus with `select` and `option` ... 342
Grouping Controls with `fieldset` and `legend` ... 350
Changing the Default Form Navigation ... 351
Using Access Keys ... 352
Creating `disabled` and `readonly` Controls ... 352
Displaying Updates with `progress` and `meter` ... 354
Applying Cascading Style Sheet Properties to Form Elements ... 359
Planning Your Forms ... 365
Summary ... 365

Workshop ... 367
Q&A ... 367
Quiz ... 368
Quiz Answers ... 368
Exercises ... 369

LESSON 13: Structuring a Page with HTML5 ... **371**

A Short History of HTML Page Layout ... 372
Laying Out a Page in HTML5 ... 373
HTML5 Structural Tags ... 375
Sections ... 375
Header ... 376
Footer ... 376
Navigation ... 377
Articles ... 378
Asides ... 379
The Page Outline ... 379
Elements with Their Own Outlines ... 381
Using HTML5 Structural Elements ... 381
Polyfill Scripts ... 381
Summary ... 381
Workshop ... 382
Q&A ... 382
Quiz ... 383
Quiz Answers ... 383
Exercise ... 383

LESSON 14: Integrating Multimedia: Video and Sound ... **385**

Embedding Video the Simple Way ... 386
Advantages and Disadvantages of Hosting Videos on External Sites ... 387
Uploading Videos to YouTube ... 388
Customizing the Video Player ... 389
Other Services ... 390
Hosting Your Own Video ... 391
Video and Container Formats ... 392
Converting Video to H.264 ... 393

Embedding Video Using <video> 396
The <video> Tag 397
Using the <source> Element 399
Embedding Flash Using the <object> Tag 400
Alternative Content for the <object> Tag 403
The <embed> Tag 404
Embedding Flash Movies Using SWFObject 406
Flash Video Players 408
JW Player 408
Using Flowplayer 410
Using the <object> Tag with the <video> Tag 412
Embedding Audio in Your Pages 413
The <audio> Tag 413
Flash Audio Players 414
Summary 416
Workshop 417
Q&A 417
Quiz 418
Quiz Answers 418
Exercises 419

LESSON 15: Advanced CSS: Page Layout in CSS 421

Laying Out the Page 422
The Problems with Layout Tables 422
Writing HTML with Structure 423
Writing a Layout Style Sheet 426
The Floated Columns Layout Technique 433
The Role of CSS in Web Design 435
Style Sheet Organization 436
Site-Wide Style Sheets 439
Summary 439
Workshop 440
Q&A 440
Quiz 441
Quiz Answers 441
Exercises 441

LESSON 16: Using Responsive Web Design 443

What Is Responsive Web Design? 444

History of Responsive Web Design 444

Why RWD Is Important 445

RWD Is More Than Just Changing the Number of Columns 445

Mobile Devices Should Come First 445

Mobile First 446

Affecting the Viewport 447

Planning a Responsive Website 449

Check Your Analytics 449

Try the Site with Your Own Phone 449

Decide What Content Is Critical 449

Writing Media Queries 450

Media Types 450

Media Features 451

Breakpoints 452

Building a Style Sheet with Media Queries 454

Understanding the Mechanics of RWD 454

Adjusting the Layout 454

Making Images and Videos Responsive 460

Building Responsive Tables 463

Responsive Web Design Best Practices 467

Give Everyone the Best Experience 467

Use the Best Breakpoints for Your Website, Not for Devices 468

Be Flexible But Think Small 468

Summary 468

Workshop 469

Q&A 469

Quiz 469

Quiz Answers 469

Exercises 470

PART IV: Using JavaScript and jQuery

LESSON 17: Introducing JavaScript 471

Why Would You Want to Use JavaScript? 472
Ease of Use 473
Improving Performance 473
Integration with the Browser 473
The `<script>` Tag 473
The Structure of a JavaScript Script 474
The `src` Attribute 474
JavaScript and the Chrome Development Tools 475
The JavaScript Language 476
Operators and Expressions 479
Variables 480
Control Structures 482
Functions 486
Data Types 488
Arrays 489
Objects 489
The JavaScript Environment 491
Events 491
Summary 495
Workshop 495
Q&A 495
Quiz 496
Quiz Answers 496
Exercises 497

LESSON 18: Using jQuery 499

What Are JavaScript Libraries? 500
Getting Started with jQuery 500
Your First jQuery Script 501
Selecting Elements from the Document 503
Binding Events 504
Modifying Styles on the Page 505
Hiding and Showing Elements 505
Retrieving and Changing Style Sheet Properties 507

Modifying Content on the Page ... 508
Manipulating Classes ... 508
Manipulating Form Values ... 511
Manipulating Attributes Directly ... 513
Adding and Removing Content ... 514
Special Effects ... 518
AJAX and jQuery ... 520
Using AJAX to Load External Data ... 521
Summary ... 525
Workshop ... 525
Q&A ... 525
Quiz ... 526
Quiz Answers ... 526
Exercises ... 526

LESSON 19: Using JavaScript in Your Pages ... **527**

Validating Forms with JavaScript ... 528
Hiding and Showing Content ... 538
The Same Code with jQuery ... 546
Adding New Content to a Page ... 546
Summary ... 551
Workshop ... 551
Q&A ... 551
Quiz ... 552
Quiz Answers ... 552
Exercises ... 553

LESSON 20: Working with Frames and Linked Windows ... **555**

What Are Frames? ... 556
Why Were Frames Removed from HTML5? ... 556
What About Iframes? ... 556
Working with Linked Windows ... 557
Browsing Context Keywords ... 557
The `<base>` Tag ... 562
Inline Frames ... 563
Opening Linked Windows with JavaScript ... 567
Summary ... 571

Workshop 572
Q&A 572
Quiz 572
Quiz Answers 572
Exercises 573

PART V: Designing for Everyone

LESSON 21: Designing for the Mobile Web 575
People Browse Differently on Mobile Phones 576
Standards Compliance and the Mobile Web 577
Progressive Enhancement 577
Validating Your Pages 578
Writing for the Mobile Web 581
Write Clearly and Be Brief 582
Organize Your Pages for Quick Scanning 582
Make Each Page Stand on Its Own 583
Be Careful with Emphasis 583
Don't Use Browser-Specific Terminology 584
Spell Check and Proofread Your Pages 584
Design and Page Layout 585
Use Headings as Headings 585
Group Related Information Visually 586
Use a Consistent Layout 586
Using Links 587
Mobile Users Tap; They Don't Click 587
Use Link Menus with Descriptive Text 587
Use Links in Text 588
Avoid the "Here" Syndrome 588
To Link or Not to Link 589
Using Images and Multimedia 591
Don't Overuse Images 591
Keep Images Small 591
Watch Out for Assumptions About Your Visitors' Hardware 592
Don't Make Your Videos Annoying 592
Avoid Flash 593

Making the Most of CSS and JavaScript 593
Put Your CSS and JavaScript in External Files 593
Location Matters 594
Shrink Your CSS and JavaScript 594
Take Advantage of Mobile Features 595
Geolocation 595
Make Phone Calls 596
SMS 597
Other Good Habits and Hints for Mobile Web Design 597
Link Back to Home 597
Don't Split Topics Across Pages 598
Sign Your Pages 598
One Final Secret to Mobile Web Design 599
Summary 599
Workshop 601
Q&A 601
Quiz 602
Quiz Answers 602
Exercises 603

LESSON 22: Designing for User Experience 605

Considering User Experience Level 606
Add a Search Engine 606
Use Concise, Sensible URLs 606
Navigation Provides Context 608
Are Your Users Tourists or Regulars? 609
Determining User Preferences 610
What Is Accessibility? 611
Common Myths Regarding Accessibility 611
Section 508 612
Alternative Browsers 613
Writing Accessible HTML 613
Tables 614
Links 615
Images and Multimedia 616
Designing for Accessibility 617
Using Color 617
Fonts 618

Take Advantage of All HTML Tags 619
Frames 619
Forms 619
Validating Your Sites for Accessibility 620
Summary 621
Workshop 621
Q&A 622
Quiz 623
Quiz Answers 623
Exercises 624

PART VI: Going Live on the Web

LESSON 23: How to Publish Your Site 625
What Does a Web Server Do? 626
Other Things Web Servers Do 626
How to Find Web Hosting 627
Using a Web Server Provided by Your School or Work 627
Using a Commercial Web Host 628
Commercial Web Builders 629
Setting Up Your Own Server 629
Free Hosting 630
Organizing Your HTML Files for Publishing 630
Questions to Ask Your Webmaster 630
Keeping Your Files Organized with Directories 632
Having a Default Index File and Correct Filenames 632
Publishing Your Files 633
Moving Files Between Systems 633
Troubleshooting 636
I Can't Access the Server 636
I Can't Access Files 636
I Can't Access Images 636
My Links Don't Work 637
My Files Are Being Displayed Incorrectly 638
Promoting Your Web Pages 638
Getting Links from Other Sites 639
Content Marketing Through Guest Posting 639

Promoting Your Site Through Social Media 640
Creating a Facebook Page for Your Site 642
Site Indexes and Search Engines 643
Business Cards, Letterhead, Brochures, and Advertisements 644
Finding Out Who's Viewing Your Web Pages 644
Log Files 644
Google Analytics 645
Summary 648
Workshop 648
Q&A 649
Quiz 649
Quiz Answers 649
Exercises 650

LESSON 24: Taking Advantage of the Server 651

How PHP Works 652
Getting PHP to Run on Your Computer 653
The PHP Language 654
Comments 654
Variables 655
Arrays 656
Strings 659
Conditional Statements 661
PHP Conditional Operators 662
Loops 662
`foreach` Loops 663
`for` Loops 664
`while` and `do...while` Loops 664
Controlling Loop Execution 665
Built-In Functions 666
User-Defined Functions 666
Returning Values 667
Processing Forms 668
Handling Parameters with Multiple Values 669
Presenting the Form 674
Using PHP Includes 678
Choosing Which Include Function to Use 680

Expanding Your Knowledge of PHP ... 681
Database Connectivity ... 682
Regular Expressions ... 682
Sending Mail ... 682
Object-Oriented PHP ... 682
Cookies and Sessions ... 683
File Uploads ... 683
Other Application Platforms ... 683
Microsoft ASP.NET ... 683
Java EE ... 684
Ruby on Rails ... 684
Summary ... 684
Workshop ... 684
Q&A ... 685
Quiz ... 685
Quiz Answers ... 686
Exercises ... 686

LESSON 25: Search Engines and SEO ... 687

What Is SEO? ... 688
Why You Need SEO ... 688
What About Social Media? ... 688
You Can Do Your Own SEO ... 689
Why Don't Search Engines Find Sites Without SEO? ... 689
How Search Engines Work ... 689
Google ... 690
Microsoft Bing ... 691
Yahoo! ... 691
Don't Forget International Searches ... 691
SEO Techniques ... 692
Is Your Site "Friendly?" ... 692
Using Keywords and Keyword Research ... 693
Creating Content for Customers Is the Best SEO ... 694
Myths and Facts About SEO ... 695
Tools for Tracking and Managing SEO ... 696
Using Sitemaps ... 696
The `robots.txt` File ... 697

Understanding Canonical Links ... 698
Redirecting Duplicate Content ... 698
Checking How Your Site Looks to Search Engines ... 699
Tracking Your SEO Efforts ... 700
Paying for Links ... 700
Summary ... 701
Workshop ... 701
Q&A ... 701
Quiz ... 702
Quiz Answers ... 702
Exercises ... 702

Index ... 703

Introduction

Over the past decade, the Web has become completely integrated into the fabric of society. Most businesses have websites, and it's unusual to see a commercial on television that doesn't display a URL. The simple fact that most people know what a URL *is* speaks volumes. People who didn't know what the Internet was several years ago are now reconnecting with their high school friends on Facebook.

Perhaps the greatest thing about the Web is that you don't have to be a big company to publish things on it. The only things you need to create your own website are a computer with access to the Internet and the willingness to learn. Obviously, the reason you're reading this is that you have an interest in web publishing. Perhaps you need to learn about it for work, or you're looking for a new means of self-expression, or you want to post baby pictures on the Web so that your relatives all over the country can stay up-to-date. The question is, how do you get started?

There's more than enough information on the Web about how to publish websites like a seasoned professional. There are tutorials, reference sites, tons of examples, and free tools to make it easier to publish on the Web. However, the advantage of reading this book instead is that all the information you need to build websites is organized in one place and presented in an orderly fashion. It has everything you need to master HTML, publish sites to a server on the Web, create graphics for use on the Web, and keep your sites running smoothly.

But wait, there's more. Other books on how to create web pages just teach you the basic technical details, such as how to produce a boldface word. In this book, you'll also learn why you should be producing a particular effect and when you should use it. In addition, this book provides hints, suggestions, and examples of how to structure your overall website, not just the words on each page. This book won't just teach you how to create a website—it'll teach you how to create a great website and how to get people to come visit it.

In this book, examples are written in valid HTML5 and CSS3 using tags that work in all current browsers wherever possible. Exceptions and caveats are noted whenever I use tags that are obsolete or not included in HTML5.

NOTE

Visit our website and register this book at www.informit.com/register for convenient access to any updates, downloads, or errata that might be available for this book.

Who Should Read This Book

Is this book for you? That depends:

- If you've seen what's out on the Web and you want to contribute your own content, this book is for you.
- If you work for a company that wants to create a website and you're not sure where to start, this book is for you.
- If you're an information developer, such as a technical writer, and you want to learn how the Web can help you present your information online, this book is for you.
- If you're just curious about how the Web works, some parts of this book are for you, although you might be able to find what you need on the Web itself.
- If you've created web pages before with text, images, and links, and you've played with a table or two and set up a few simple forms, you may be able to skim the first half of the book. The second half should still offer you a lot of helpful information.

What This Book Contains

The lessons are arranged in a logical order, taking you from the simplest tasks to more advanced techniques:

- Part I: Getting Started

  In Part I, you'll get a general overview of the World Wide Web and what you can do with it. You'll also write your first (basic) web page with HTML and CSS.

- Part II: Creating Web Pages

  In Part II, you'll learn how to write simple documents in the HTML language and style them with CSS. You'll learn how to create lists on your pages as well as paragraphs of text, and you'll learn how to link your pages with hypertext links.

- Part III: Doing More with HTML and CSS

  In Part III, you'll learn the meat of building web pages. You'll learn how to format text and style a page using CSS. You'll learn how to add images and create tables and forms and place them on your pages. You'll also learn how to lay out your web pages with CSS and make them responsive to the devices that are viewing them.

- Part IV: Using JavaScript and jQuery

  In Part IV, you'll learn how you can extend the functionality of your web pages by adding JavaScript to them. First, we provide an overview of JavaScript and of jQuery. We provide some specific JavaScript examples you can use on your own pages. And you learn how to make inline frames and linked windows.

- Part V: Designing for Everyone

  Part V gives you hints for creating a well-constructed website, and you'll learn how to design for mobile devices as well as make your site accessible so that it is usable by people with disabilities.

- Part VI: Going Live on the Web

  In Part VI, you'll learn how to put your site up on the Web, including how to advertise the work you've done. You'll also learn how to use some of the features of your web server to make your life easier. And you'll get some tips for making your site searchable in the most popular search engines with search engine optimization (SEO).

What You Need Before You Start

There are lots of books about how to use the Web. This book isn't one of them. We're assuming that if you're reading this book, you already have a working connection to the Internet, you have a modern web browser such as Chrome, Safari, Firefox, Opera, Internet Explorer version 10, or Microsoft Edge, and that you're familiar with the basics of how the Web and the Internet work. You should also have at least a passing acquaintance with some other elements of the Internet, such as email and FTP, because we refer to them in general terms in this book.

In other words, you need to have used the Web to provide content for the Web. If you meet this one simple qualification, read on!

Many of the screenshots in this book are made on a Macintosh computer, but you can do all the work on Windows or a Linux machine if that's what you use. You should just be familiar with how your operating system works and where common programs are located.

Conventions Used in This Book

This book uses special typefaces and other graphical elements to highlight different types of information.

Special Elements

Three types of "boxed" elements present pertinent information that relates to the topic being discussed: Note, Tip, and Caution as follows:

NOTE

Notes highlight special details about the current topic.

TIP

It's a good idea to read the tips because they present shortcuts or trouble-saving ideas for performing specific tasks.

CAUTION

Don't skip the cautions. They help you avoid making bad decisions or performing actions that can cause you trouble.

▼ Task

Tasks demonstrate how you can put the information in a lesson into practice by giving you a real working example. ▲

HTML Input and Output Examples

Throughout the book, we present exercises and examples of HTML input and output.

Input ▼

An input icon identifies HTML code that you can type in yourself.

Output ▼

An output icon indicates the results of the HTML input in a web browser such as Microsoft Internet Explorer.

Special Fonts

Several items are presented in a monospace font, which can be plain or italic. Here's what each one means:

- `plain mono`—Applied to commands, filenames, file extensions, directory names, and HTML input. For example, HTML tags such as `<table>` and `<p>` appear in this font.
- *`mono italic`*—Applied to placeholders. A placeholder is a generic item that replaces something specific as part of a command or computer output. For instance, the term represented by *`filename`* would be the real name of the file, such as `myfile.txt`.

Workshop

In the "Workshop" section, you can reinforce your knowledge of the concepts in the lesson by answering quiz questions or working on exercises. The Q&A provides additional information that didn't fit in neatly elsewhere in the lesson.

LESSON 1
What Is Web Publishing?

A journey of a thousand miles begins with a single step, and here you are in Lesson 1 of a journey that will show you how to write, design, and publish pages on the World Wide Web. But before beginning the actual journey, you should start simple, with the basics. You'll learn the following:

- How the World Wide Web really works
- What web browsers do, and which browsers your audience will be using
- What a web server is, and why you need one
- Some information about *uniform resource locators* (URLs)

These days, the Web is pervasive, and today's lesson might seem like old news. If so, feel free to skim it and skip ahead to Lesson 2, "Getting Your Tools in Order," where you'll discover the first steps you need to take to learn to create web pages.

Thinking Like a Web Publisher

You're almost certainly already familiar with the Web as a user. You open your favorite web browser and visit websites where you look up information, shop, or keep up with what your friends are doing. You may also use your web browser to read your email, check your calendar, and do your work.

Being a web publisher means understanding what happens when you enter an address in your web browser or click a link and visit a website. But first, before I get into explaining the Web at a technical level, I want to define it at a conceptual level.

The Web is

- A hypertext information system
- Cross-platform
- Distributed
- Dynamic
- Interactive

So, let's look at all these words and see what they mean in the context of how you use the Web as a publishing medium.

The Web Is a Hypertext Information System

The idea behind hypertext is that instead of reading text in a rigid, linear structure (such as a book), you can skip easily from one point to another. You can get more information, go back, jump to other topics, and navigate through the text based on what interests you at the time.

Hypertext enables you to read and navigate text and visual information in a nonlinear way, based on what you want to know next.

When you hear the term *hypertext*, think *links*. (In fact, some people still refer to links as hyperlinks.) Whenever you visit a web page, you're almost certain to see links throughout the page. Some of the links might point to locations within that same page, others to pages on the same site, and still others might point to pages on other sites. Hypertext was an old concept when the Web was invented—it was found in applications such as HyperCard and various help systems. However, the World Wide Web redefined how large a hypertext system could be. Even large websites were hypertext systems of a scale not before seen, and when you take into account that it's no more difficult to link to a document on a server in Australia from a server in the United States than it is to link to a document stored in the same directory, the scope of the Web becomes truly staggering.

NOTE

Nearly all large corporations and medium-sized businesses and organizations are using web technology to manage projects, order materials, and distribute company information in a paperless environment. By locating their documents on a private, secure web server called an *intranet*, they take advantage of the technologies the World Wide Web has to offer while keeping the information contained within the company.

The Web Is Cross-Platform

If you can access the Internet, you can access the World Wide Web, regardless of whether you're working on a smartphone, a tablet, a brand new laptop, or a desktop computer you bought at the flea market. If you think Windows menus and buttons look better than Macintosh menus and buttons or vice versa (or if you think both Macintosh and Windows people are weenies), it doesn't matter. The World Wide Web isn't limited to any one kind of machine or developed by any one company. The Web is entirely cross-platform.

Cross-platform means that you can access web information equally well from any computer hardware running any operating system using any display.

The Cross-Platform Ideal

The whole idea that the Web is—and should be—cross-platform is strongly held to by purists. The reality, however, is somewhat different. With the introduction over the years of numerous special features, technologies, and media types, the cross-platform nature of the Web has been compromised. Web authors can choose to use nonstandard features, like Flash, but in doing so they limit the potential audience for their site, especially as more and more people switch to smartphones and mobile devices to view the Web. Web publishers also must choose between creating native applications for mobile devices or using modern web standards to build web applications that are more cross-platform compatible. It's up to individual creators to decide whether to compromise cross-platform flexibility for the greater capabilities of a proprietary platform.

The Web Is Distributed

Web content can take up a great deal of storage, particularly when you include images, audio, and video. To store all the information published on the Web, you would need

an untold amount of disk space, and managing it would be almost impossible. (Not that there aren't people who try.)

The Web succeeds at providing so much information because that information is distributed globally across millions of websites, each of which contributes the space for the information it publishes. These sites reside on one or more computers, referred to as web servers. A *web server* is just a computer that listens for requests from web browsers and responds to that request. You, as a consumer of that information, request a resource from the server to view it. You don't have to install it or do anything other than point your browser at that site.

A *website* is a location on the Web that publishes some kind of information. When you view a web page, your browser connects to that website to get that information.

Each website, and each page or bit of information on that site, has a unique address. This address is called a *uniform resource locator* or URL. When people tell you to visit a site at http://www.nytimes.com/, they've just given you a URL. Whenever you use a browser to visit a website, you get there using a URL. You'll learn more about URLs later in this lesson in the "Uniform Resource Locators" section.

The Web Is Dynamic

If you want a permanent copy of some information that's stored on the Web, you have to save it locally because the content can change any time, even while you're viewing the page.

If you're browsing that information, you don't have to install a new version of the help system, buy another book, or call technical support to get updated information. Just launch your browser and check out what's there.

If you're publishing on the Web, you can make sure that your information is up-to-date all the time. You don't have to spend a lot of time re-releasing updated documents. There's no cost of materials. You don't have to get bids on numbers of copies or quality of output. Color is free. And you won't get calls from hapless customers who have a version of the book that was obsolete four years ago.

Consider a book published and distributed entirely online, such as *Little Brother* by Cory Doctorow (which you can find at http://craphound.com/littlebrother/). He can correct any mistakes in the book and simply upload the revised text to his website, making it instantly available to his readers. He can post pointers to foreign language translations of the book as they arrive. The website for the book appears in Figure 1.1.

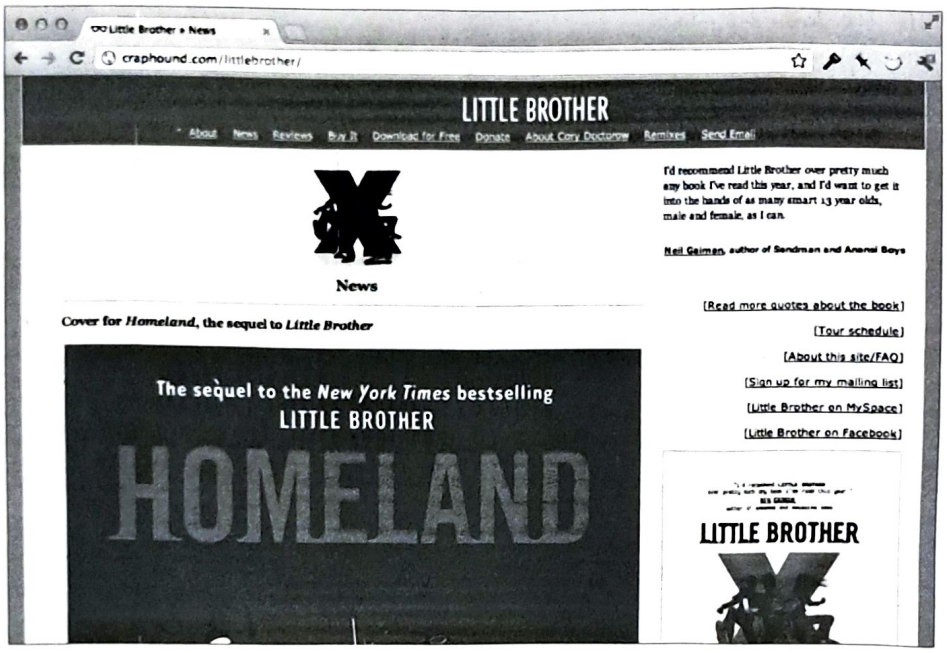


FIGURE 1.1
The website for *Little Brother*.

NOTE

The pictures throughout this book are taken in Google Chrome or Safari running on OS X. The only reason for this use is that I'm writing this book on an Apple Macintosh. If you're using a different operating system, don't feel left out. As I noted earlier, the glory of the Web is that you see the same information regardless of the platform you're using.

For some sites, the capability to update the site on-the-fly, at any moment, is precisely why the site exists. Figure 1.2 shows the home page for the BBC News, a site that's updated 24 hours a day to reflect up-to-the-minute news as it happens. Because the site is up and available all the time, it has an immediacy that newspapers cannot match. Visit the BBC News at http://www.bbc.co.uk/news/world/.

These days, you don't even need to reload a web page to receive updated information. Through the use of JavaScript, which I discuss starting in Lesson 17, "Introducing JavaScript," you can update the contents of a page in real time. The scores and statistics on the NBA game page in Figure 1.3 are updated in place as the game progresses.

FIGURE 1.2
The BBC News.

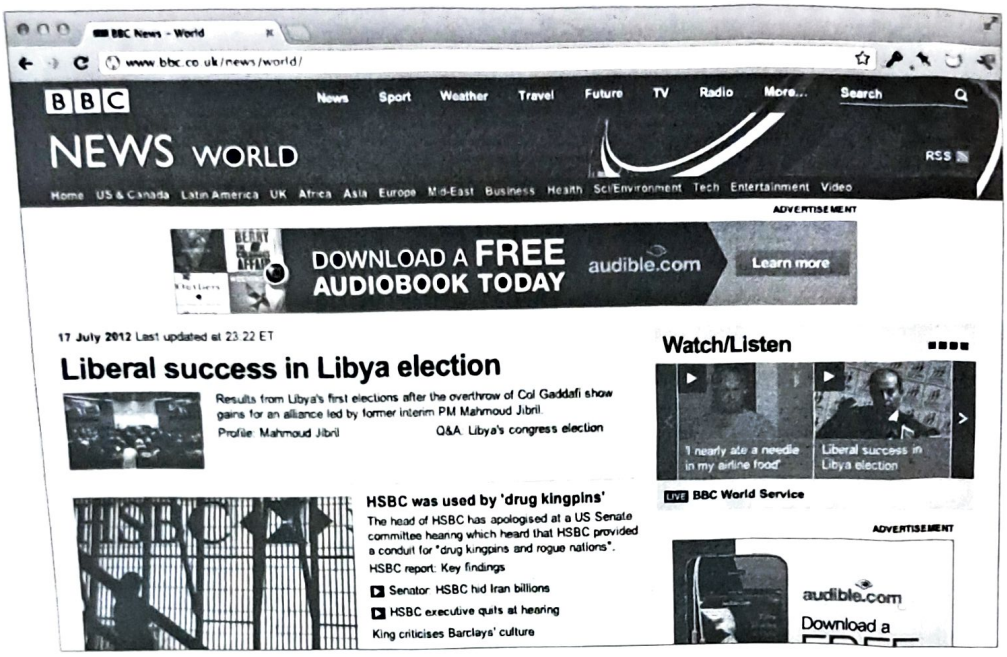


FIGURE 1.3
Live game updates on the CBS Sports website.



The Web Is Interactive

Interactivity is the capability to "talk back" to the web server. More traditional media, such as television, isn't interactive in the slightest; all you do is sit and watch as shows are played at you. Other than changing the channel, you don't have much control over

what you see. The Web is inherently interactive; the act of selecting a link and jumping to another web page to go somewhere else on the Web is a form of interactivity. In addition to this simple interactivity, however, the Web enables you to communicate with the publisher of the pages you're reading and with other readers of those pages.

Indeed, the most popular sites on the Web these days are about interacting with other users of the site rather than with the site's publisher. That's what people mean when they say "social media." Rather than spending money to hire writers and cameramen, now sites are spending money to hire programmers to create spaces for people to share content they create with one another. These days, it's not uncommon to see people on TV reading viewer posts from Twitter or Facebook out loud on the air. Such is the degree to which this form of media has taken hold.

As a web publisher, you'll need to decide the type of interaction you want your site to provide. You can publish web pages without any outlet for users to interact. You can enable users to submit feedback privately. You can enable them to publish public comments and converse with you and with each other. You can provide forums that enable users to interact with one another directly. You can provide games or other interactive features. You can even incorporate interactive features from other websites into your own so that you can integrate your site with the sites to which your users already belong. For example, Figure 1.4 shows a Facebook widget incorporated into a third-party website.

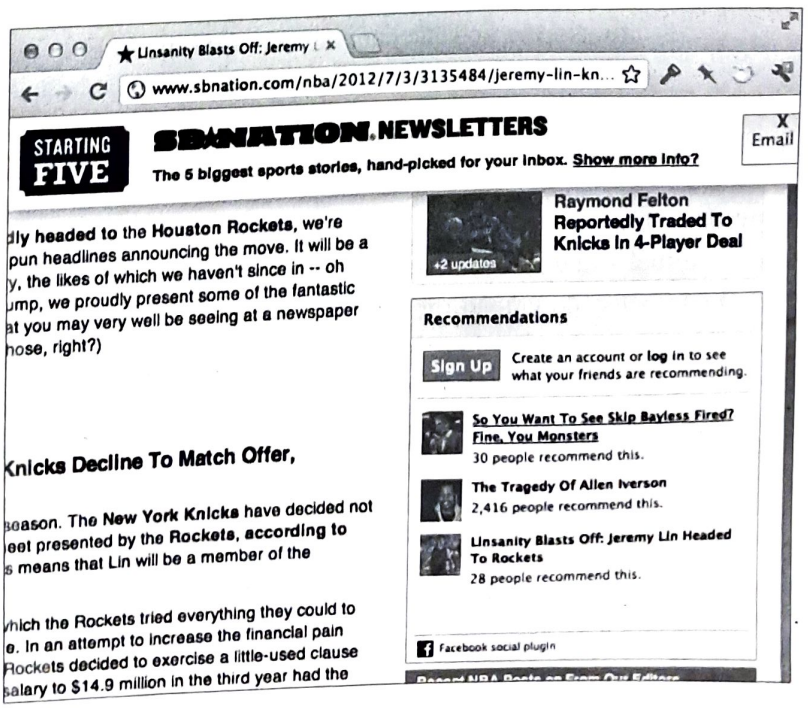


FIGURE 1.4
A Facebook widget.

Web Browsers

A web browser, as mentioned earlier, is the application you use to view pages and navigate the World Wide Web. A wide array of web browsers is available for just about every platform you can imagine. Microsoft Internet Explorer, for example, is included with Windows, and Safari is included with OS X. Mozilla Firefox, Google Chrome, and Opera are all available as free downloads. Likewise, more and more people are using browsers on mobile devices and tablets. iPhone and iPad use Mobile Safari. The Android mobile platform has its own browser. There are also other mobile platforms, like BlackBerry and Windows Phone, and third-party browsers for both Android and iPhone. Not too many years ago, Internet Explorer was the dominant browser for Windows, the OS X market share was less than 5%, and mobile browsers were so limited that they wouldn't work with regular web pages at all. Back then, developers sometimes chose to support Internet Explorer and ignore other browsers. That is no longer a viable strategy.

NOTE

Choosing to develop for a specific browser, such as Internet Explorer, is only suitable when you know a limited audience using the targeted browser software will view your website. Developing this way is a common practice in corporations implementing intranets. In these situations, it's a fair assumption that all users in the organization will use the browser supplied to them and, accordingly, it's possible to design the web pages on an intranet to use the specific capabilities of the browser in question.

What the Browser Does

The core purpose of a web browser is to connect to web servers, request documents, and then properly format and display those documents. Web browsers can also display files on your local computer, download files that are not meant to be displayed, and in some cases even allow you to send and retrieve email. What the browser is best at, however, is retrieving and displaying web documents. Each web page is written in a language called the *Hypertext Markup Language* (HTML) that includes the text of the page, a description of its structure, and links to other documents, images, or other media. The browser takes the information it gets from the web server and formats it for your display. Different browsers might format and display the same file in diverse ways, depending on the capabilities of that system and how the browser is configured.

An Overview of Some Popular Browsers

There's a good chance you use only one browser, or two, if you use a browser on a computer and one on a mobile device. However, your website will probably be visited by a variety of browsers, and to publish on the Web successfully, you'll need to be aware of them. This section describes some of the most popular browsers on the Web. They're in no way the only browsers available, and if the browser you're using isn't listed here, don't feel that you have to use one of these. Whichever browser you have is fine as long as it works for you.

Google Chrome

Google Chrome is currently the most popular web browser. Its market share has shown incredible growth because the browser offers great performance and stability and is updated often, plus it is used on both desktop and mobile devices. It uses the same HTML engine as Apple's Safari browser, an open source engine called WebKit. It's available as a free download at http://www.google.com/chrome/. You'll see Google Chrome used for the screenshots in this book, and I'll be talking about its special features for people creating websites, starting in Lesson 2.

Microsoft Internet Explorer

Microsoft's browser, Microsoft Internet Explorer, is included with Microsoft Windows and is still the second most popular web browser. It has lost market share to other browsers because new versions are not released as often as Google Chrome and Mozilla Firefox. However, a huge number of people still use Internet Explorer, and it is the most unlike other browsers like Chrome, Firefox, and Safari. According to the website CanIUse.com, Internet Explorer 10 offers 49% support of HTML5 features, and Internet Explorer 11 offers 58% support.

NOTE

If you're serious about web design, you should install all the popular browsers on your system and use them to view your pages after you've published them. That way, you can make sure that everything is working properly. Even if you don't use a particular browser on a day-to-day basis, your site will be visited by people who do. If you are interested in checking cross-browser compatibility issues, start with Microsoft Internet Explorer and Mozilla Firefox, and include Google Chrome, too.

Figure 1.5 shows Microsoft Edge—the successor to Internet Explorer—running under Windows 10.

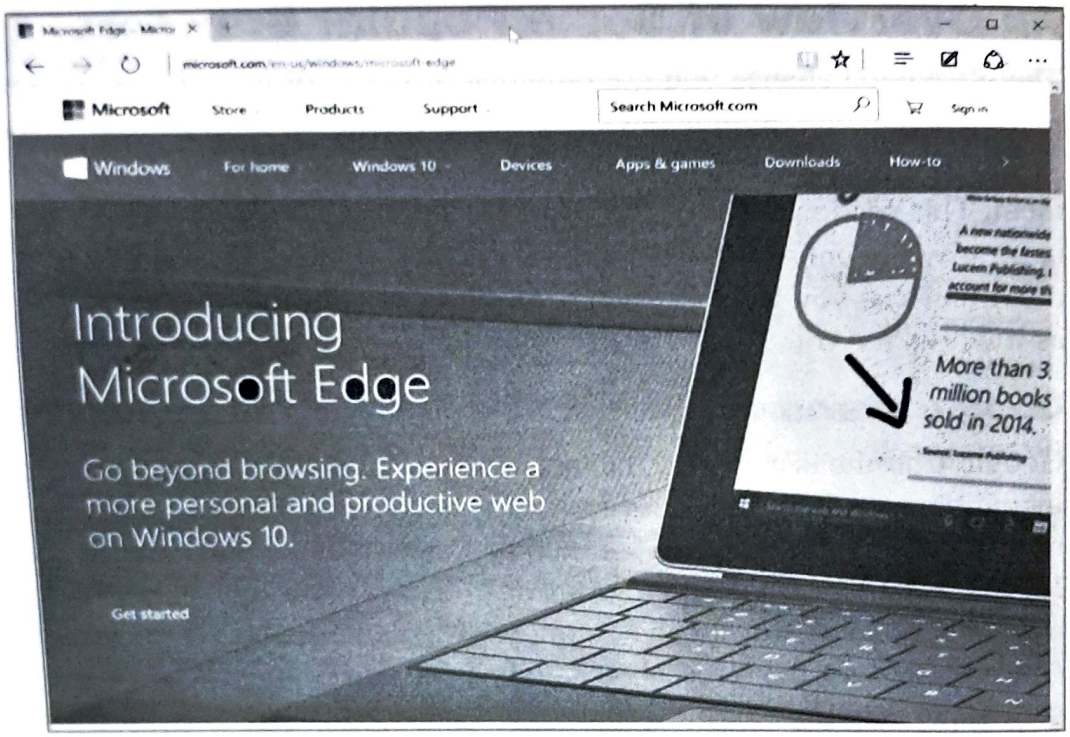


FIGURE 1.5 Microsoft Edge (Windows 10).

One other important point to make about Internet Explorer is that the different versions differ greatly. Version 10 of Internet Explorer was released in 2012, but many users haven't upgraded from version 9, version 8, or even version 7. Internet Explorer differs widely between versions, so to get a site to work properly you need to test in each version. Web publishers have dropped support for version 6, and most have also dropped version 7, and Microsoft recommends that all users upgrade to a newer version. And in 2015 Microsoft released a new browser Microsoft Edge—the default browser for Windows 10.

Mozilla Firefox

Mozilla Firefox is a free, open source web browser that makes up roughly 15% of the browser market as of July 2015. Netscape Navigator was the first popular commercial web browser. Version 1.0 was released in 1994. In 1998, Netscape Communications opened the source code to their web browser and assigned some staff members to work on making it better. Seven years and many releases later, the result of that effort was Mozilla Firefox. Netscape Communications, since acquired by America Online, no longer has any official ties to the Mozilla Foundation, which is now an independent nonprofit organization.

Firefox became popular in large part because it was free from the security issues that plagued Internet Explorer. In addition, a large number of Firefox extensions improve the browser experience, and Firefox has done a good job of keeping up with web standards as they have evolved. Firefox is available for Windows, Mac OS X, and Linux and is a free download at http://www.mozilla.com/.

Apple Safari

Safari is the default browser for OS X. There is also a mobile version of this browser installed on the Apple iPhone and iPad. It is based on open source technology, and its support for web standards is at a similar level to Firefox. Right now, Safari has around 9% of the browser market share.

Mobile Browsers

No discussion of browsers would be complete without talking about mobile browsers. The big three are Chrome, Safari, and Android. As of July 2015, Chrome has 31% of the market share for mobile browsers, with Safari and Android at 24% and 15%, respectively.

Google Chrome came on the market for mobile devices before 2014 and became the most popular browser on mobile devices in early 2015. It uses the same engine as the desktop version of the browser and offers the performance and reliability people have come to expect of Chrome. Safari is the browser Apple includes with iOS devices like the iPhone and iPad. It offers very strong HTML5 support and, apart from screen size, provides an experience very similar to a desktop browser. Similarly, Android provides a browser that also provides a high-quality web experience. All three of them are based on the WebKit rendering engine, just like Safari and Chrome for the desktop. I'll discuss the considerations that go into building sites that are friendly to mobile devices in Lesson 21, "Designing for the Mobile Web."

Other Browsers

As of July 2015, Google Chrome has the lion's share of the market for web browsers on both desktop and mobile devices. The remaining browsers all share a relatively small slice of the pie—13% or less. For example, Opera (http://www.operasoftware.com/) has a niche market with only 5% share. It's small, fast, free, and available for a number of platforms, including Windows, Mac OS X, and Linux. It's also standards compliant. For UNIX users who use KDE, there's Konqueror. There are various Mozilla offshoots, such as Camino for Mac OS X. Likewise, command-line browsers such as Lynx and Links are available to provide an all-text view of web pages. There are also a number of browsers that provide access to the Web for people with various special needs. It makes sense to code to common standards to accommodate all these types of browsers.

Web Servers

To view and browse pages on the Web, all you need is a web browser. To publish pages on the Web, you need a web server.

A *web server* is the program that runs on a computer and is responsible for replying to web browser requests for whatever content is associated with a particular URL. You need a web server to publish documents on the Web. One point of confusion is that the computer on which a server program runs is also referred to as a server. So, when someone uses the term *web server*, she could be referring to a program used to respond to requests for web pages or the computer on which that program runs.

When you use a browser to request a page on a website, that browser makes a web connection to a server using HTTP. The server accepts the connection, sends the contents of the requested files, and then closes the connection. The browser then formats the information it got from the server.

On the server side, many different browsers can connect to the same server to get the same information. The web server is responsible for handling all these requests.

Web servers do more than just serve files. They're also responsible for managing form input and for linking forms and browsers with programs such as databases running on the server.

As with browsers, many different servers are available for many different platforms, each with many different features. For now, all you need to know is what the server is there for; you'll learn more about web servers in Lesson 23, "How to Publish Your Site."

Uniform Resource Locators

As you learned earlier, a URL is a pointer to some bit of data on the Web, be it a web document, an image, a style sheet, or a JavaScript script. You'll learn about all of these later. The URL provides a universal, consistent method for finding and accessing information.

In addition to typing URLs directly into your browser to go to a particular page, you also use URLs when you create a hypertext link within a document to another document. So, any way you look at it, URLs are important to how you and your browser get around on the Web.

URLs contain information about the following:

- How to get to the information (which protocol to use: FTP, HTTP, or file)
- The Internet hostname of the computer where the content is stored (www.ncsa.uiuc.edu, ftp.apple.com, netcom16.netcom.com, and so on)
- The directory or other location on that site where the content is located

You also can use special URLs for tasks such as sending mail to people (called *Mailto URLs*) and running JavaScript code. You'll learn all about URLs and what each part means in Lesson 6, "Working with Links."

Defining Web Publishing Broadly

When the Web was invented, web publishing meant one thing: creating web pages as individual files and uploading them to a server so that people could view them in their browsers. Since then, pretty much everything has changed.

A few websites still include hand-coded web pages that the creator uploads, but most websites are created using software that runs on the server. Web pages have gotten more complex, as have websites. These days, most content on web pages is written using applications that live on the Web as well. For example, you can create a blog at WordPress.com and immediately begin posting content through the WordPress web interface.

Whether you're posting status updates on Twitter, writing comments on a news site, publishing a blog through a tool, or editing articles on Wikipedia, you're publishing on the Web. In most cases, you are not required to directly write HTML on your own. Generally, the pages live in templates that someone else created, and often you can format the content you create using a graphical editor or with simplified markup that enables you to avoid the use of HTML.

Ultimately the content, however you enter it, will be converted to HTML before it is displayed to users. So if you publish something and it doesn't look right, you'll need to know HTML if you want to fix it. You'll need to be able to differentiate between the parts of the page you control and the parts that are built in to the publishing application that you're using. And if you want to take greater control of the appearance of your site, you will probably need to know HTML to update the templates that are used to give your pages their own look and feel.

So no matter what approach you take to web publishing, you will likely benefit by starting with the basics and learning how web publishing works from end to end. You may never write individual web pages by hand, but understanding how to do so will prepare you to build websites using whichever tool you ultimately choose.

Summary

To publish on the Web, you have to understand the basic concepts that make up the parts of the Web. In this lesson, you learned three major concepts. First, you learned about a few of the more useful features of the Web for publishing information. Second, you learned about web browsers and servers and how they interact to deliver web pages. Third, you learned about what a URL is and why it's important to web browsing and publishing.

Workshop

Each lesson in this book contains a workshop to help you review the topics you learned. The first section of this workshop lists some common questions about the Web. Next, you'll answer some questions that I'll ask you about the Web. The answers to the quiz appear in the next section. At the end of each lesson, you'll find some exercises that will help you retain the information you learned about the Web.

Q&A

Q Who runs the Web? Who controls all these protocols? Who's in charge of all this?

A No single entity owns or controls the World Wide Web. Given the enormous number of independent sites that supply information to the Web, for any single organization to set rules or guidelines would be impossible. Two groups of organizations, however, have a great influence over the look and feel and direction of the Web itself.

The first is the *World Wide Web Consortium* (W3C), based at Massachusetts Institute of Technology in the United States and INRIA in Europe. The W3C is made up of individuals and organizations interested in supporting and defining the languages and protocols that make up the Web (HTTP, HTML, XHTML, and so on). It also provides products (browsers, servers, and so on) that are freely available to anyone who wants to use them. The W3 Consortium is the closest anyone gets to setting the standards for and enforcing rules about the World Wide Web. You can visit the Consortium's home page at http://www.w3.org/.

The second group of organizations that influences the Web is the browser developers themselves, most notably Google, Apple, Microsoft, and the Mozilla Foundation. The competition to be the most popular and technically advanced browser on the Web can be fierce. A group of people and companies interested in the future of the Web have created an organization called the Web Hypertext Application Technology Working Group (or WHATWG). The WHATWG, along with the W3C, wrote the HTML5 specification.

Going forward, the WHATWG has abandoned version numbers for the HTML specification entirely. Instead, HTML will be a "living standard" and incorporate both experimental and widely supported features. The goal is to make sure that the specification evolves to match the features that browser makers have agreed to add to their browsers. If a proposed feature does not reach consensus, it is removed from the specification. This is an attempt to prevent the problems of the past where the process of creating the HTML specification diverged from the work the browser makers were doing.

Q I've heard that the Web changes so fast that it's almost impossible to stay current. Is this book doomed to be out-of-date the day it's published?

A Although it's true that things do change on the Web, the vast majority of the information in this book will serve you well far into the future. HTML is as stable now as it has ever been, and once you learn the core technologies of *Hypertext Markup Language* (HTML), *Cascading Style Sheets* (CSS), and JavaScript, you can add on other things at your leisure.

Quiz

1. What's a URL?
2. What's required to publish documents on the Web?

Quiz Answers

1. A URL, or uniform resource locator, is an address that points to a specific document or bit of information on the Internet.
2. You need access to a web server. Web servers, which are programs that serve up documents over the Web, reply to web browser requests for files and send the requested pages to many different types of browsers. They also manage form input and handle database integration.

Exercises

1. Start thinking more about web publishing as you surf the Web. Look at how URLs are constructed. Pay attention to how the pages are constructed. Soon you'll understand how these pages are built from the inside out.
2. Download a different browser than the one you ordinarily use and try it out for a while. If you're using Internet Explorer, try out Firefox, Chrome, Safari, or even a command-line browser such as Lynx or Links. To really see how things have changed and how some users who don't upgrade their browser experience the Web, download an old browser from http://browsers.evolt.org/ and try it out.